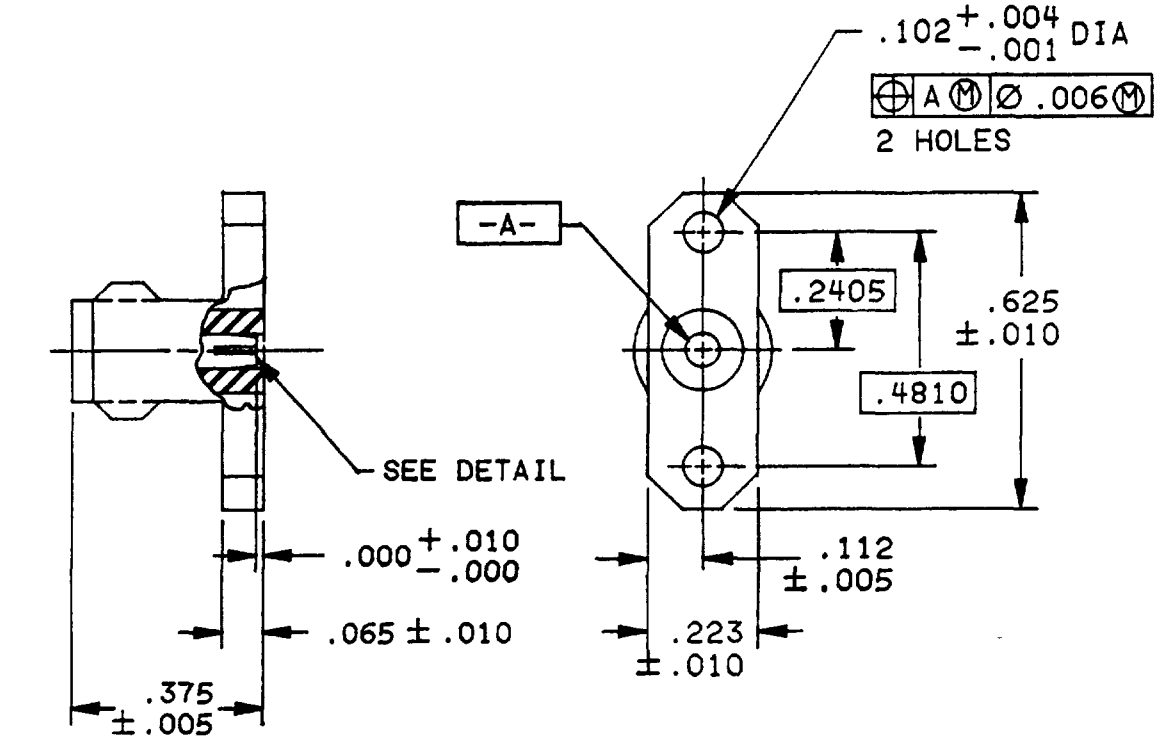
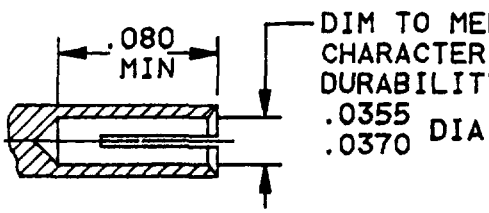
MILITARY SPECIFICATION SHEET

CONNECTOR, COAXIAL, RADIO FREQUENCY, STRIP OR MICROSTRIP TRANSMISSION LINE, SERIES SMA (SOCKET CONTACT, FLANGE MOUNTED RECEPTACLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-83517.





DETAIL

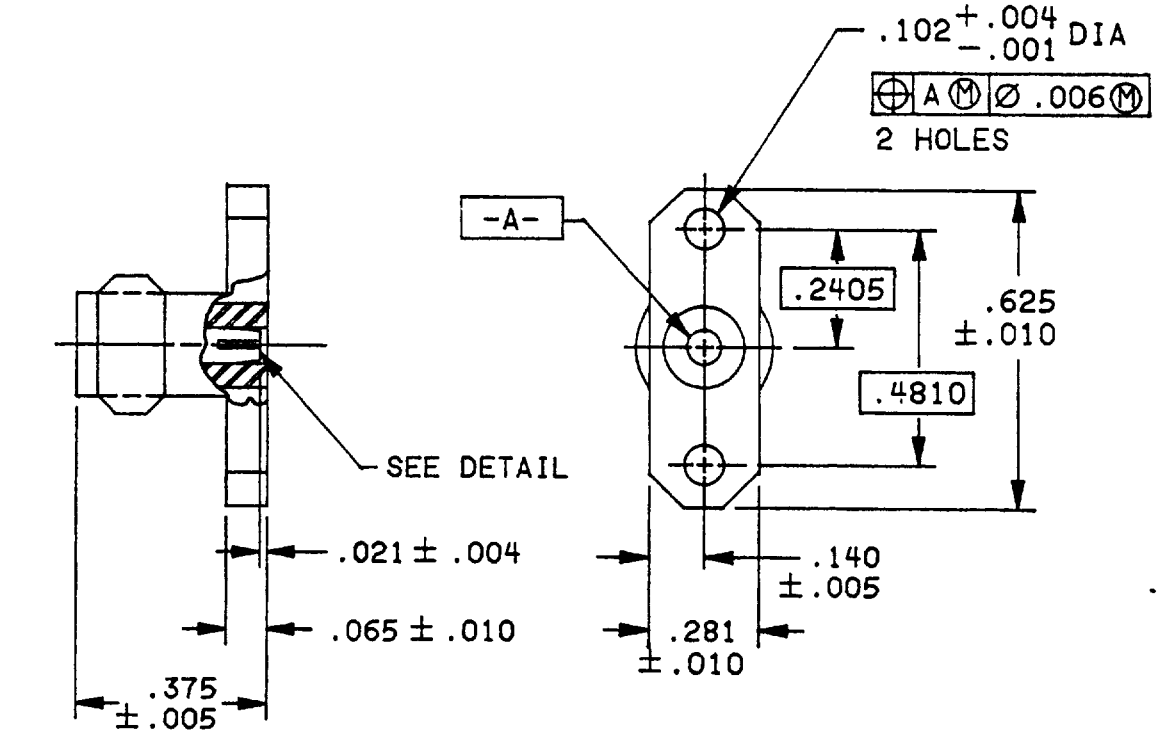
INCHES	MM	INCHES	MM
.001	.03	.0370	.940
.004	.10	.080	2.03
.005	.13	.102	2.59
.006	.15	.112	2.84
.010	.25	.223	5.66
.0355	.902	.2405	6.11
		.375	12.22

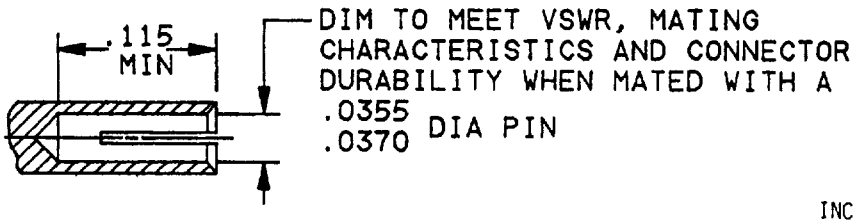
NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Number two screws are suggested for mounting.

FIGURE 1. Series SMA socket contact, 2 hole (.223 inch wide) flange mounted receptacle.

FSC 5935

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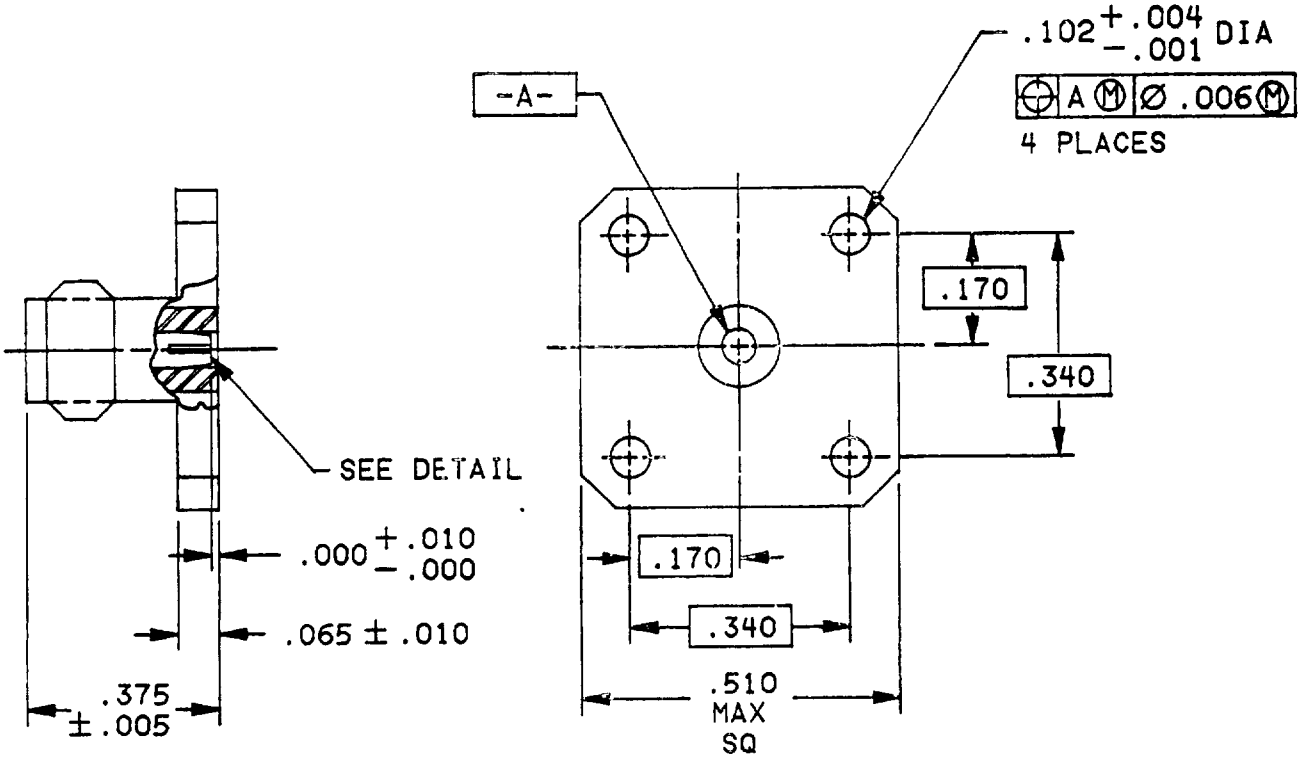
DETAIL

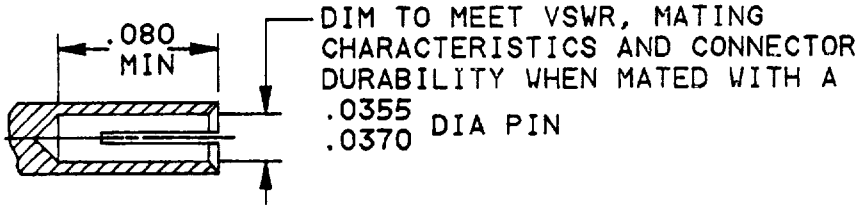
INCHES	MM	INCHES	MM
.001	.03	.065	1.65
.004	.10	.102	2.59
.005	.13	.115	2.92
.006	.15	.140	3.56
.010	.25	.2405	6.11
.0355	.90	.375	9.53
.0370	.94	.4810	12.22
		.625	15.88

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Number two screws are suggested for mounting.

FIGURE 2. Series SMA, socket contact 2 hole (.281 inch wide) flange mounted receptacle.

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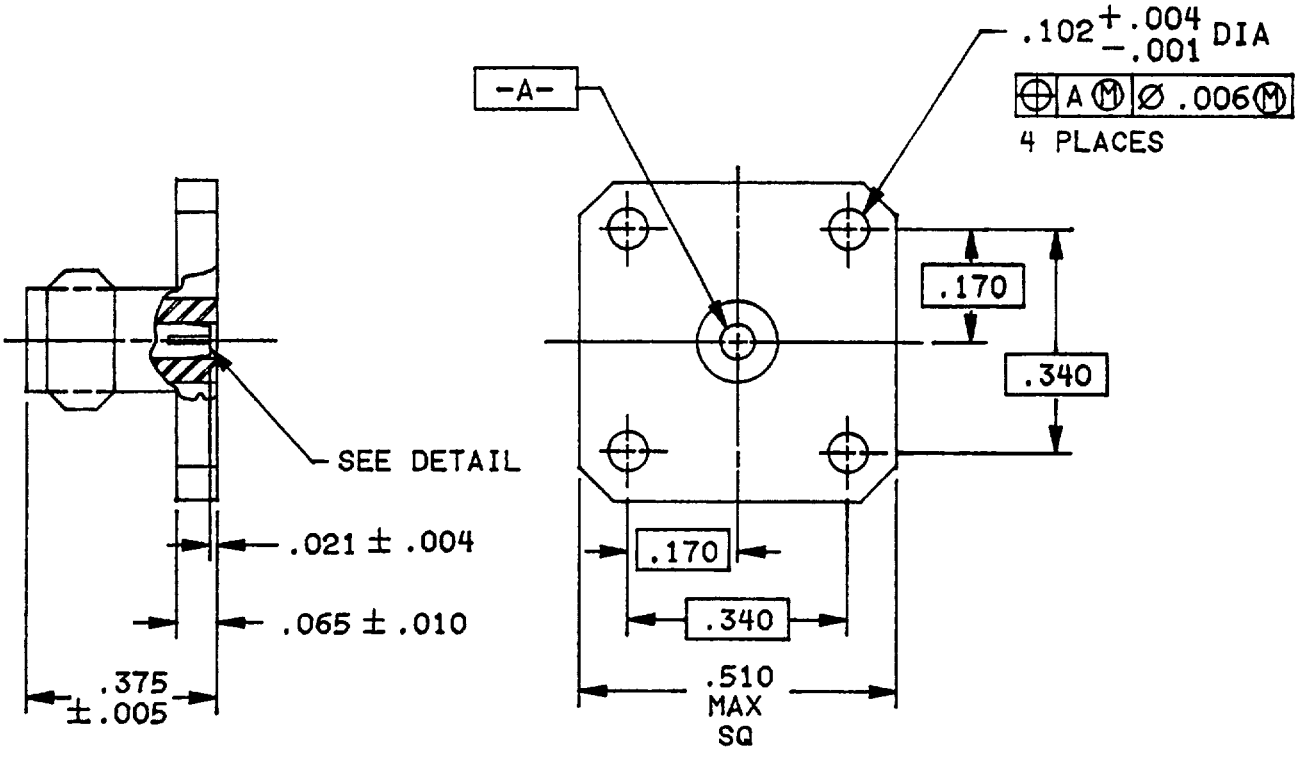
DETAIL

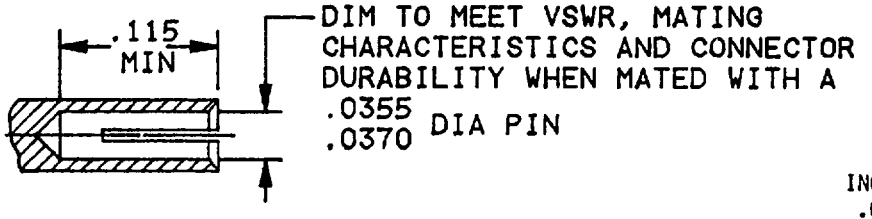
INCHES	MM	INCHES	MM
.001	.03	.065	1.65
.004	.10	.080	2.03
.005	.13	.102	2.59
.006	.15	.170	4.32
.010	.25	.340	8.64
.0355	.90	.375	9.53
.0370	.940	.510	12.95

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Number two screws are suggested for mounting.

FIGURE 3. Series SMA (.080 inch minimum) socket contact (4 hole) flange mounted receptacle.

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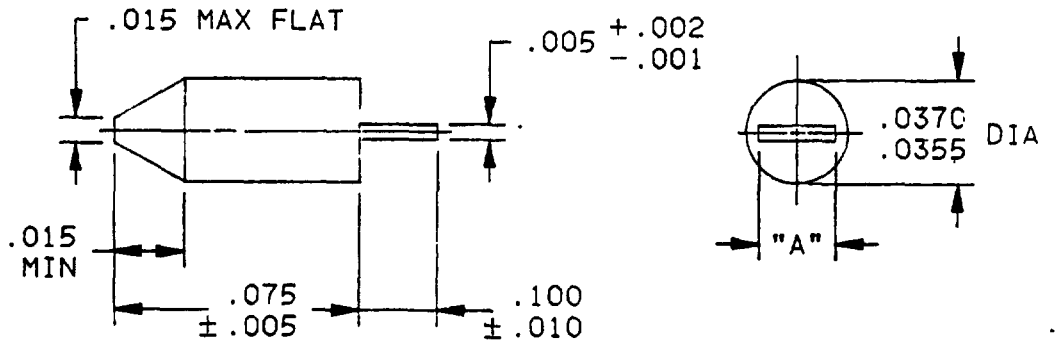
DETAIL

INCHES	MM	INCHES	MM
.001	.03	.0370	.940
.004	.10	.065	1.65
.005	.13	.102	2.59
.006	.15	.115	2.92
.010	.25	.170	4.32
.021	.53	.340	8.64
.0355	.90	.375	9.53
		.510	12.95

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Number two screws are suggested for mounting.

FIGURE 4. Series SMA (.115 inch minimum) socket contact (4 hole) flange mounted receptacle.

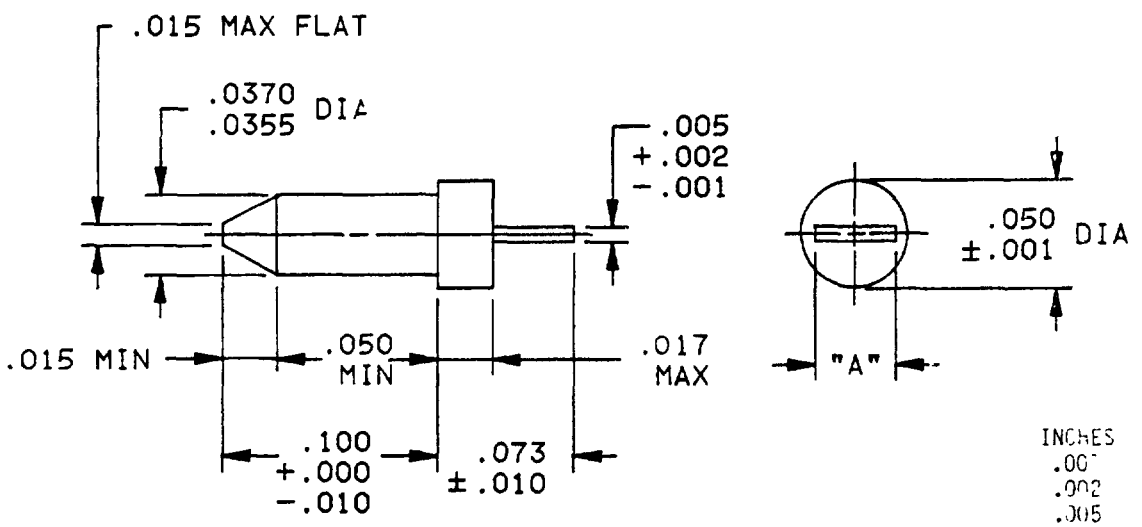
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* FIGURE	"A" DIM
5 A	.020 ±.002
5 B	.050 ±.002

INCHES	MM
.001	.03
.002	.05
.005	.13
.010	.25
.015	.38
020	.51
.0355	.90
.0370	.94
.050	1.27
.075	1.10
.100	2.54

FIGURE 5. Transition pin (.075 inch).



* FIGURE	"A" DIM
6 A	.020 ±.002
6 B	.050 ±.001

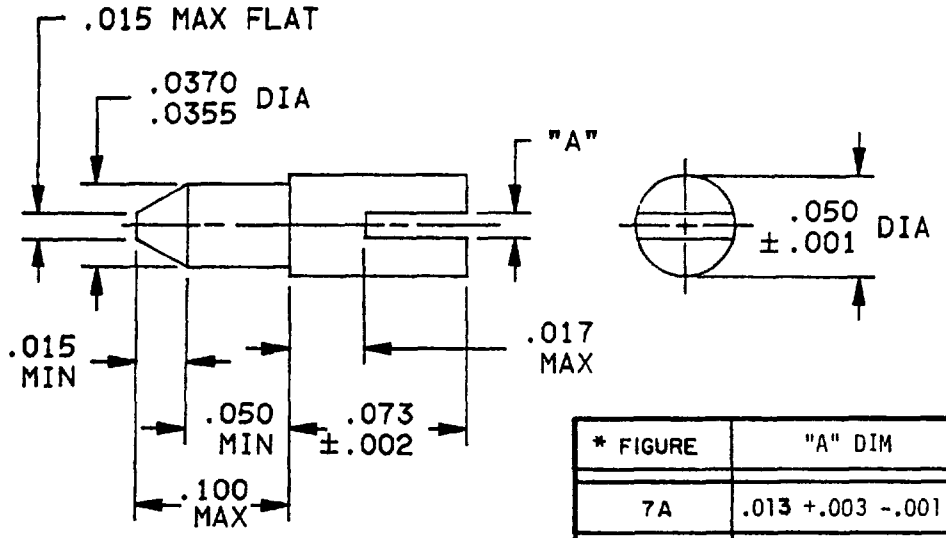
INCHES	MM
.001	.03
.002	.05
.005	.13
.010	.25
.015	.38
.017	.43
.020	.51
.0355	.901
.0370	.940
.050	1.27
.073	1.85
.100	2.54

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. All unidmensioned pictorial configurations are for reference purposes only.

* 4 See Table 1

FIGURE 6. Transition pin (.100 inch).

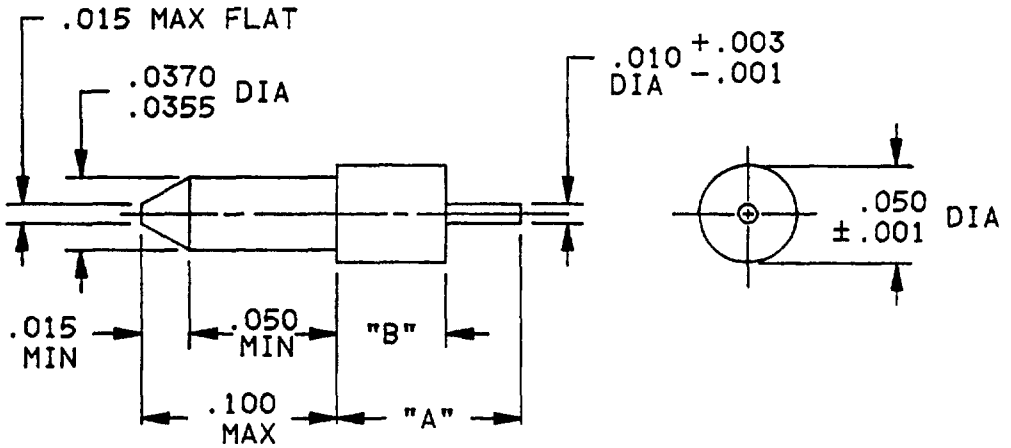
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INCHES	MM
.001	.03
.002	.05
.003	.08
.012	.30
.015	.38
.017	.43
.018	.46
.028	.71
.0355	.901
.0377	.940
.050	1.27
.073	1.85
.100	2.54

* FIGURE	"A" DIM
7A	.013 +.003 -.001
7B	.018 +.003 -.001
7C	.028 +.003 -.001

FIGURE 7. Transition pin (slotted).



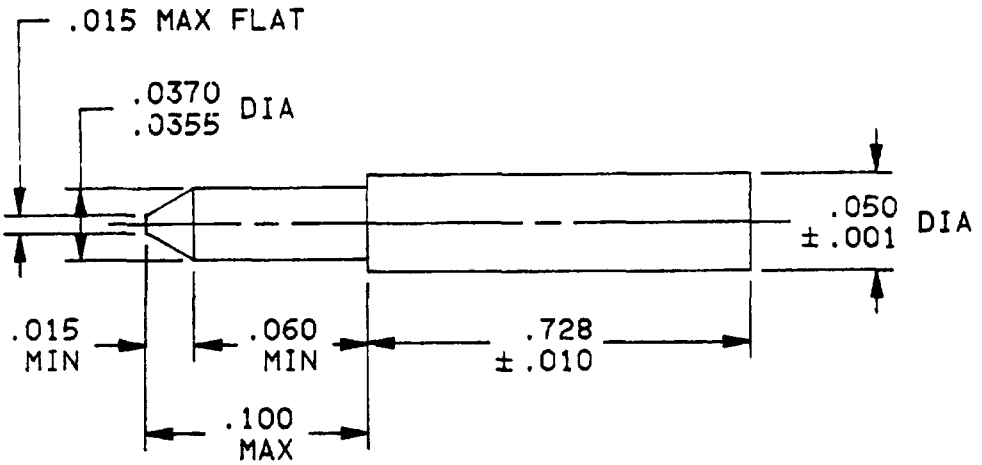
INCHES	MM
.001	.03
.002	.05
.003	.08
.010	.25
.015	.38
.0355	.901
.0370	.940
.050	1.27
.080	2.03
.100	2.54
.136	3.45
.142	3.61
.198	5.03

* FIGURE	"A" DIM	"B" DIM
8A	.136 ±.010	.080 ±.002
8B	.198 ±.010	.142 ±.002
8C	073 ± 010	.015 ± 002

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.
* 4. See Table I.

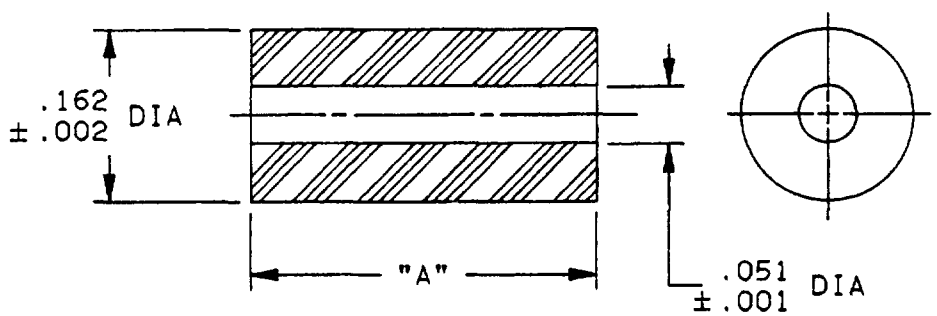
FIGURE 8. Transition pin (.050 inch contact).

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INCHES	MM
.001	.03
.002	.05
.005	.13
.010	.25
.015	.38
.020	.51
.0355	.90
.0370	.94
.050	1.27
.075	1.90
.100	2.54

FIGURE 9. Transition Pin (.060 inch contact).



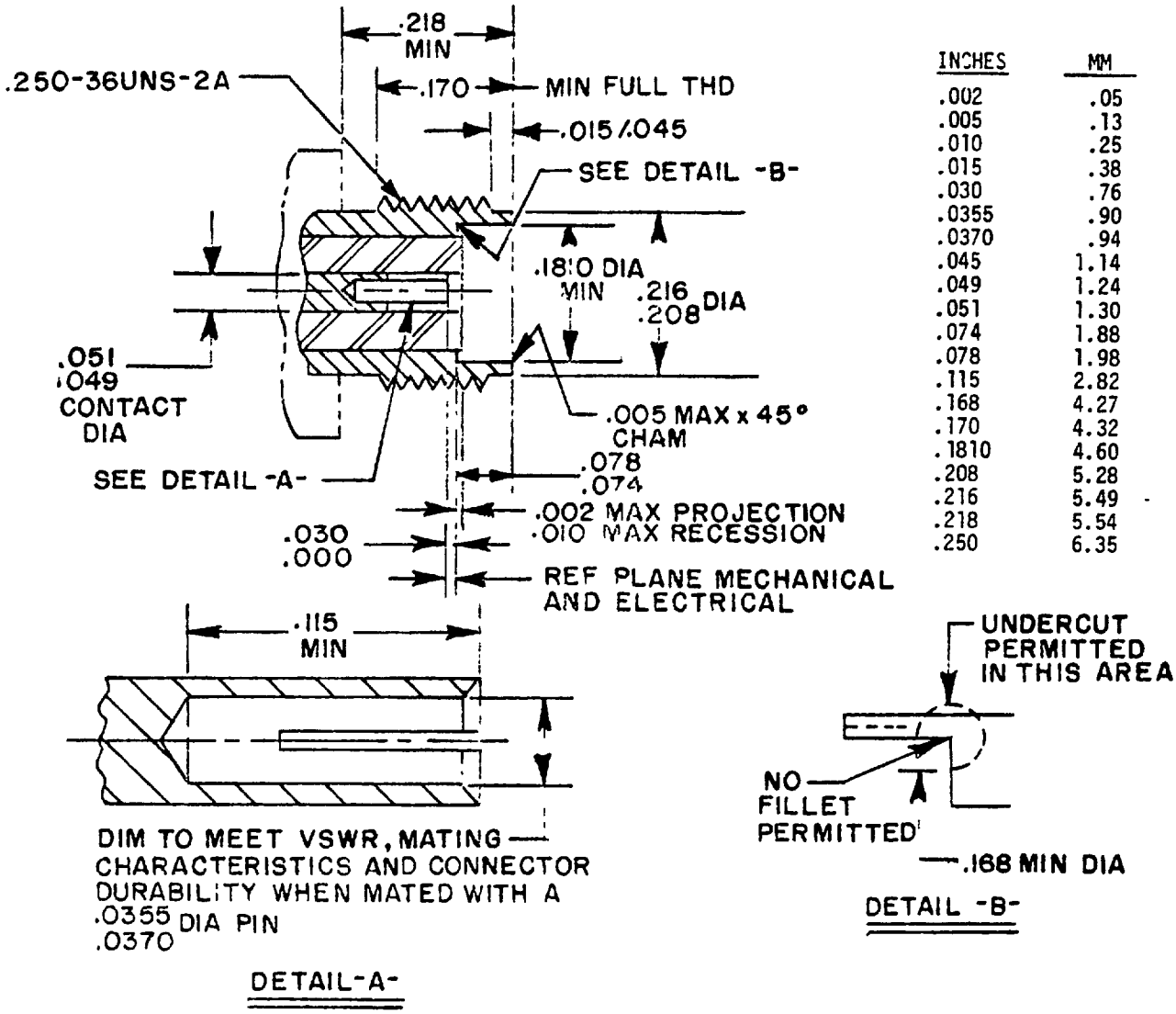
* FIGURE	"A" DIM
10 A	.070 ±.002
10 B	.132 ± 002
10 C	.625 ±.002

INCHES	MM	INCHES	MM
.001	.03	.070	1.78
.002	.05	.132	3.35
.051	1.30	.162	4.11
		.625	15.88

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.
* 4. See Table 1.

FIGURE 10. Rear dielectric.

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INCHES	MM
.002	.05
.005	.13
.010	.25
.015	.38
.030	.76
.0355	.90
.0370	.94
.045	1.14
.049	1.24
.051	1.30
.074	1.88
.078	1.98
.115	2.82
.168	4.27
.170	4.32
.1810	4.60
.208	5.28
.216	5.49
.218	5.54
.250	6.35

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only

FIGURE 11. Mating dimensions for socket termination.

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ENGINEERING PARAMETERS

Nominal impedance: 50 ohms.

Voltage rating: 600 Vrms maximum at sea level.

150 Vrms maximum at 70,000 feet.

Frequency range: 0 to 18.0 GHz.

Temperature rating: -65° to 105°C.

REQUIREMENTS.

Design and construction. See figures 1 through 11 and table I.

Force to engage and disengage:

Torque - 2 inch-pounds maximum.

Longitudinal force - Not applicable.

Coupling proof torque: Not applicable.

Inspection note: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Contact gaging. See figure 12.

Contacts with spring members:

Center contact (socket)

Oversize test pin - .0375 +.0001

Test pin finish - 16 microinches.

Insertion depth - .030/.045.

Number of insertions - 3.

Insertion force test: Steel test pin diameter .0370 +.0001.

Insertion depth - .050/.075.

Test pin finish - 16 microinches.

Insertion force - 3 pounds maximum.

Withdrawal force test. Steel test pin diameter .0355 - .0001.

Insertion depth - .050/.075.

Withdrawal force - 1 ounce minimum.

Test pin finish - 16 microinches.

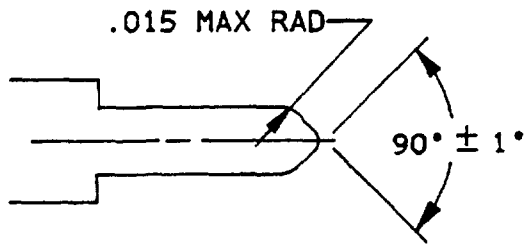


FIGURE 12. Test pin data.

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TABLE I. Part number and characteristics.

Part no. M83517/10-	Basic connector figure no.	Transition pin figure no.	Rear dielectric figure no.	Remarks
31001	1	----	---	---
31002	2	----	---	---
31003	3	----	---	---
31004	4	----	---	---
31005	1	5A	---	.020 wide tab
31006	1	5B	---	.050 wide tab
31007	2	6A	---	.020 wide tab
31008	2	6B	---	.050 wide tab
31009	2	7A	---	.012 slot
31010	2	7B	---	.018 slot
31011	2	7C	---	.028 slot
31012	2	8A	10A	.010 dia. terminal for 1/16 panel
31013	2	8B	10B	.010 dia. terminal for 1/8 panel
31014	2	9	10C	.050 dia. terminal
31015	3	5A	---	.020 wide tab
31016	3	5B	---	.050 wide tab
31017	4	6A	---	.020 wide tab
31018	4	6B	---	.050 wide tab
31019	4	7A	---	.012 slot
31020	4	7B	---	.018 slot
31021	4	7C	---	.028 slot
31022	4	8A	10A	.010 dia. terminal for 1/16 panel
31023	4	8B	10B	.010 dia. terminal for 1/8 panel
31024	4	9	10C	.050 dia. terminal

Permeability of nonmagnetic materials. Applicable.

Seal:

Hermetic-sealed connectors. Not applicable.

Pressurized and weatherproof connectors Not applicable.

Insulation resistance. 5,000 megohms minimum.

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Center contact retention.

Axial force: 6 pounds minimum.

Torque. Not applicable.

Dielectric withstanding voltage. Applicable, test condition I.

Test voltage 1,000 Vrms.

Corrosion. Applicable, test condition B.

Voltage standing wave ratio (VSWR)

Test frequency range: From .5 to 18.0 GHz.

Swept frequency VSWR test setup.

Step 1: See basic specification.

Step 2. VSWR shall be less than 1.080 +.005 frequency (frequency in GHz).

Item 11p VSWR shall be less than 1.025 + .002 frequency (frequency in GHz).

Item 11j VSWR shall be less than 1.025 + .002 frequency (frequency in GHz).

Step 3: VSWR shall be less than 1.05 + .005

Test fixture - See figure 1 of basic specification.

Step 4 VSWR test shall be less than 1.05 + .005

RF transmission loss: 0.15dB MAX @ 6 GHz

RF leakage. Not applicable.

Connector durability.

Interface

500 cycles minimum at 12 cycles/minute maximum rate.

Connector shall meet contact gaging and force to engage and disengage requirements.

Transition pin

100 cycles minimum at 12 cycles/minute maximum rate.

Contact shall meet contact gaging requirements.

Contact resistance. In milliohms maximum.

	Initial	After environment
Center contact.	6.0	8.0
Outer contact.	2.0	Not applicable

Thermal shock. Applicable, test condition A.

Moisture resistance Method 106 of MIL-STD-202.

No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

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RF high potential withstanding voltage:

At a frequency between 5 to 7.5 MHz.

Leakage current - Not applicable.

RF voltage - 1,000 Vrms.

Coupling mechanism retention force: Not applicable.

Part number: M83517/10-(dash number from table I.)

Group qualification: See table II.

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors 1/2/	Qualifies the following connectors
I	M83517/10-3†005	M83517/10-3†001 3†003 3†005 3†006 3†015 3†016
II	M83517/10-3†007	M83517/10-3†002 3†004 3†007 3†008 3†017 3†018
III	M83517/10-3†009	M83517/10-3†002 3†004 3†009 3†010 3†011 3†019 3†020 3†021
IV	M83517/10-3†012	M83517/10-3†002 3†004 3†012 3†013 3†022 3†023
V	M83517/10-3†014	M83517/10-3†002 3†004 3†014 3†024

1/ Individual connectors other than listed are self qualifying only.

2/ Qualification of connectors qualifies connectors of the same material only.

† Denotes finish.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, AT
Navy - SH, AS
Air Force - 11, 99
DLA - ES

User activities:
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 85

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